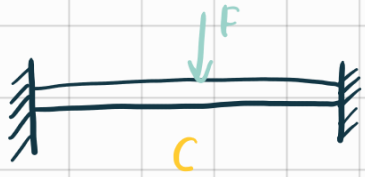
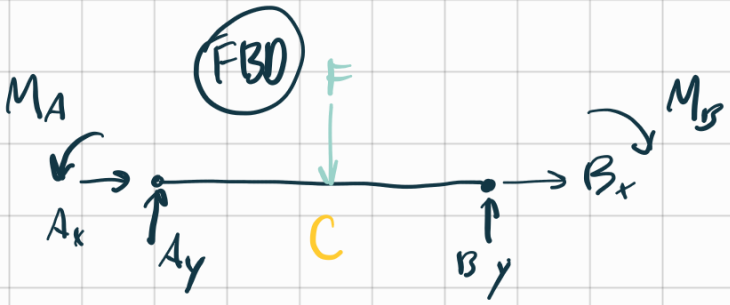
Part 1:

Error %

$$err = \frac{|FEA - Theo|}{|Theo|} \times 100$$

a)





~~6 reactions~~

4 reactions

b) $\Sigma F_y = 0$: $A_y - F + B_y = 0$ [B_x and A_x are 0]

$\circlearrowleft_+ \Sigma M_A = 0$: $M_A - F(L/2) + B_y(L) - M_B = 0$

$2(\#Joints) = \#bars + \#Reactions$

$2(2) = 1 + 4$

$4 \neq 5$ Not Determinate

c) The fixed supports will not allow for any deflection or bending.

d) $\Sigma F_y = 0$: $A_y - F + B_y = 0$

$A_y + B_y = F$ → by symetry $A_y = B_y$

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$$A_y = B_y = \frac{F}{2}$$

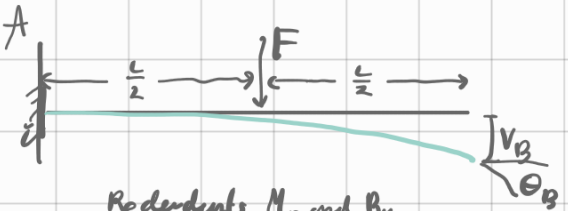
(a)



Actual Beam

+

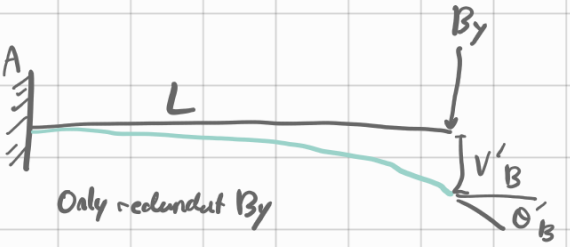
(b)



Redundants M_B and B_y removed

+

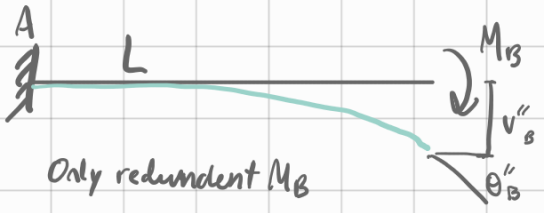
(c)



Only redundut B_y

+

(d)



Only redundent M_B

$$\theta_B + \theta'_B + \theta''_B = 0$$

$$V_B + V'_B + V''_B = 0$$

$$\theta_B = \frac{-FL^2}{8EI}$$

$$V_B = -\frac{5FL^3}{48EI}$$

$$\theta'_B = \frac{-B_y L^2}{2EI}$$

$$V'_B = \frac{-B_y L^3}{3EI}$$

$$\theta''_B = \frac{M_B L}{EI}$$

$$V''_B = \frac{M_B L^2}{2EI}$$

B_y is positive $\frac{F}{2}$

$$0 = -\frac{FL^2}{8EI} + \frac{\frac{F}{2}L^2}{2EI} + \frac{M_B L}{EI}$$

$$0 = -\frac{5FL^3}{48EI} + \frac{\frac{F}{2}L^3}{3EI} + \frac{M_B L^2}{2EI}$$

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$$0 = \frac{FL^2}{8EI} + \frac{\frac{1}{2}L}{2EI} + \frac{M_B}{EI}$$

$\times 8EI$ $\quad 0 = -\frac{FL^2}{8EI} + \frac{2FL^2}{8EI} + \frac{8M_BL}{8EI}$ $\times 8EI$

$$0 = -FL^2 + 2FL^2 + 8M_BL$$

$$0 = FL^2 + 8M_BL$$

$$-FL^2 = 8M_BL$$

$$\boxed{M_B = \frac{-FL}{8}}$$

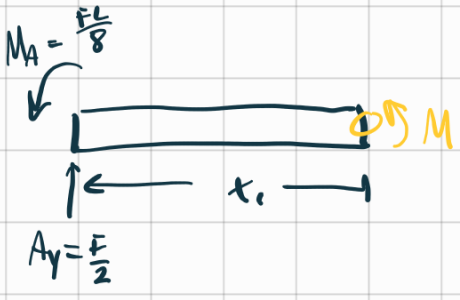
And by Symetry

$\rightarrow$ $M_A = -M_B$, so $M_A = \frac{FL}{8}$

Reaction Forces And Moments

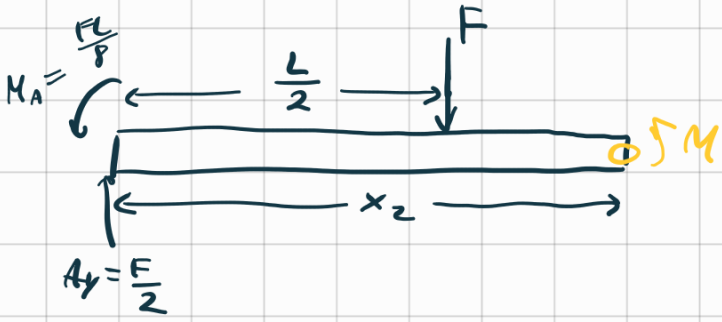
$A_y = \frac{F}{2}\uparrow$ $\quad B_y = \frac{F}{2}\uparrow$ $\quad \curvearrowleft M_A = \frac{FL}{8}$ $\quad \curvearrowright M_B = -\frac{FL}{8}$

e)



$\curvearrowleft +\Sigma M_0 = 0$: $\quad \frac{FL}{8} - \frac{F}{2}x_1 + M = 0$

$$M_1 = \frac{F}{2}x_1 - \frac{FL}{8}$$

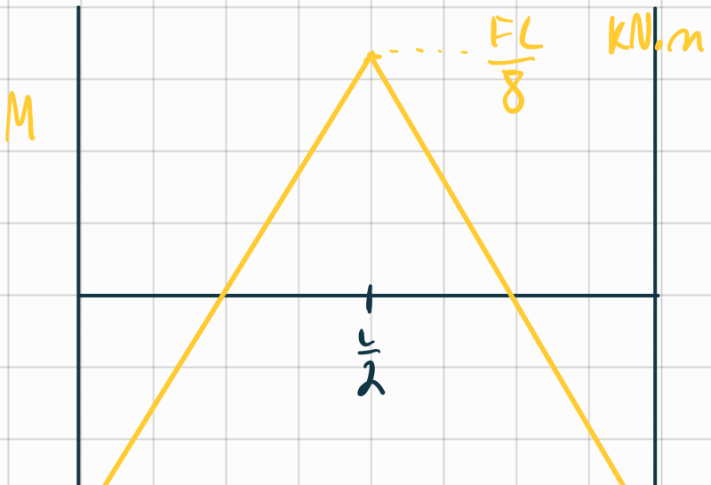


$\curvearrowleft +M_0 = 0$: $\quad \frac{FL}{8} - \frac{F}{2}x_2 + F\left(x_2 - \frac{L}{2}\right) + M = 0$

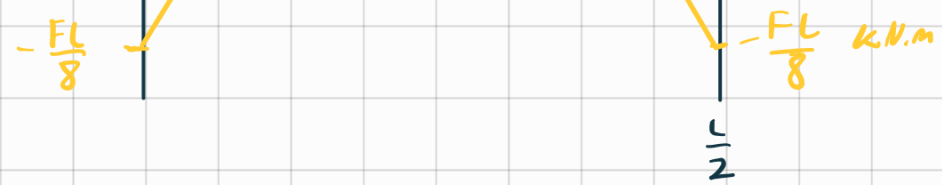
$$\frac{FL}{8} - \frac{F}{2}x_2 + Fx_2 - \frac{FL}{2} + M = 0$$

$$\frac{F}{2}x_2 - \frac{3FL}{8} + M = 0$$

$$M_2 = \frac{3FL}{8} - \frac{F}{2}x_2$$



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f) $\sigma = \frac{My}{I}$ $\quad M_{max} = \frac{FL}{8}$ $\quad$ radius $c = \left(\frac{d_o}{2}\right)$

$$\sigma_{MAX} = \frac{\frac{FL}{8}\, c}{I}$$

$$I = \frac{1}{4}\pi\left(\frac{d_o}{2}\right)^2$$

$$I = \frac{1}{16}\pi d_o^2$$

$$\boxed{\sigma_{MAX} = \frac{FL\, d_o}{16 I}}$$

g) $M_1 = \frac{F}{2}x_1 - \frac{FL}{8}$ $\quad M_2 = -\frac{F}{2}x_2 + \frac{3FL}{8}$

$EIv'' = \frac{F}{2}x_1 - \frac{FL}{8}$ $\quad EIv'' = -\frac{F}{2}x_2 + \frac{3FL}{8}$

$EIv_1' = \frac{F}{4}x_1^2 - \frac{FL}{8}x_1 + C_1$ $\quad EIv' = -\frac{Fx_2^2}{4} + \frac{3FL}{8}x_2 + C_3$

$EIv_1 = \frac{F}{12}x_1^3 - \frac{FL}{16}x_1^2 + C_1x_1 + C_2$ $\quad EIv = -\frac{Fx_2^3}{12} + \frac{3FL}{16}x_2^2 + C_3x_2 + C_4$

Boundary Conditions

@ $x=0$ $v' = v = 0$

$C_1 = 0$

$C_2 = 0$

@ $x = L$ $v = v' = 0$

$$0 = -\frac{F(L)^2}{4} + \frac{3FL}{8}(L) + C_3$$

$$C_3 = \frac{2FL^2}{8} - \frac{3FL^2}{8}$$

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$$C_3 = -\frac{FL^2}{8}$$

$$0 = -\frac{FL^3}{12} + \frac{3FL}{16}(L)^2 - \frac{FL^2}{8}(L) + C_4$$

$$C_4 = \frac{FL^3}{12} - \frac{3FL^3}{16} + \frac{FL^3}{8}$$

$$C_4 = \frac{4FL^3}{48} - \frac{9FL^3}{48} + \frac{6FL^3}{48}$$

$$C_4 = \frac{FL^3}{48}$$

Continuity Check

@ $x = \frac{L}{2}$ $\quad v_1 = v_2$ and $v_1' = v_2'$

(v') $\quad \frac{F}{4}x_1^2 - \frac{FL}{8}x_1 = -\frac{Fx_2^2}{4} + \frac{3FL}{8}x_2 - \frac{FL^2}{8}$

at $x = \frac{L}{2}$ $\quad \frac{F}{4}\left(\frac{L}{2}\right)^2 - \frac{FL}{8}\left(\frac{L}{2}\right) = -\frac{F\left(\frac{L}{2}\right)^2}{4} + \frac{3FL}{8}\left(\frac{L}{2}\right) - \frac{FL^2}{8}$

$$\frac{FL^2}{16} - \frac{FL^2}{16} = -\frac{FL^2}{16} + \frac{3FL^2}{16} - \frac{2FL^2}{16}$$

$$0 = 0 \quad \checkmark$$

(v) $\quad \frac{F}{12}x_1^3 - \frac{FL}{16}x_1^2 = -\frac{Fx_2^3}{12} + \frac{3FL}{16}x_2^2 - \frac{FL^2}{8}x_2 + \frac{FL^3}{48}$

at $x = \frac{L}{2}$ $\quad \frac{F}{12}\left(\frac{L}{2}\right)^3 - \frac{FL}{16}\left(\frac{L}{2}\right)^2 = -\frac{F\left(\frac{L}{2}\right)^3}{12} + \frac{3FL\left(\frac{L}{2}\right)^2}{16} - \frac{FL^2\left(\frac{L}{2}\right)}{8} + \frac{FL^3}{48}$

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$$\frac{FL^3}{96} - \frac{FL^3}{64} = \frac{-FL^3}{96} + \frac{3FL^3}{64} - \frac{FL^3}{16} + \frac{FL^3}{48}$$

$$\frac{FL^3}{96} - \frac{FL^3}{64} = -\frac{FL^3}{96} + \frac{2FL^3}{96} + \frac{3FL^3}{64} - \frac{4FL^3}{64}$$

$$\frac{FL^3}{96} - \frac{FL^3}{64} = \frac{FL^3}{96} - \frac{FL^3}{64} \checkmark$$

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h) $\sigma_{max} = \frac{F L d_o}{16 I}$ $\quad d_o = 25 = 0.025$ $\quad I = \frac{\pi(d_o^4 - d_i^4)}{64} = \frac{\pi(.025^4 - .017^4)}{64}$

$L = 0.77$ $\quad F = 1300N$

$$\frac{1300(0.77)(0.025)}{16\left(\frac{\pi(0.025^4 - 0.017^4)}{64}\right)} = 103752538.6 Pa = 103.75 MPa$$

i) $EIv_1 = \frac{F}{12}x_1^3 - \frac{FL}{16}x_1^2 + C_1 + C_2$, where $C_1 = C_2 = 0$

$$v_1 = \frac{\frac{F}{12}\left(\frac{L}{2}\right)^3 - \frac{FL}{16}\left(\frac{L}{2}\right)^2}{EI} = \frac{\frac{1300}{12}\left(\frac{0.77}{2}\right)^3 - \frac{1300}{16}\left(\frac{0.77}{2}\right)^2}{210\cdot10^9 \cdot \left(\frac{\pi(.025^4 - .017^4)}{64}\right)}$$

$= -9.76 \cdot 10^{-4} m = -0.976 mm$

Note: Max deflection happens at $\frac{L}{2}$

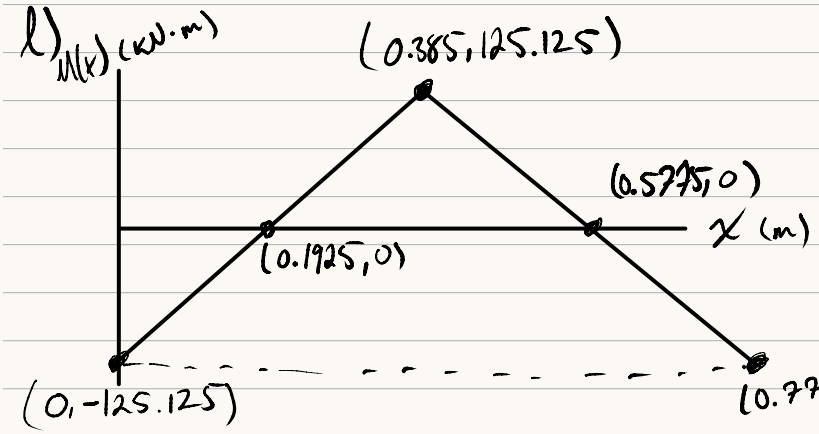
j)

	h + i	Deliverable 1+2	Deliverable 3+4	Deliverable 5+6
Max normal stress (MPa)	103.75	88.5	88.5	88.5
Max deflection (mm)	-0.976	-0.88	-0.885	-0.91

Info from the ME10 Lab graphs

k) The FEA analysis underestimates the actual amount of stress. The possible cause for this is that the model does not include the finer details of the object. Therefore, the weight distribution will be different, making the center lighter than it should.

l)



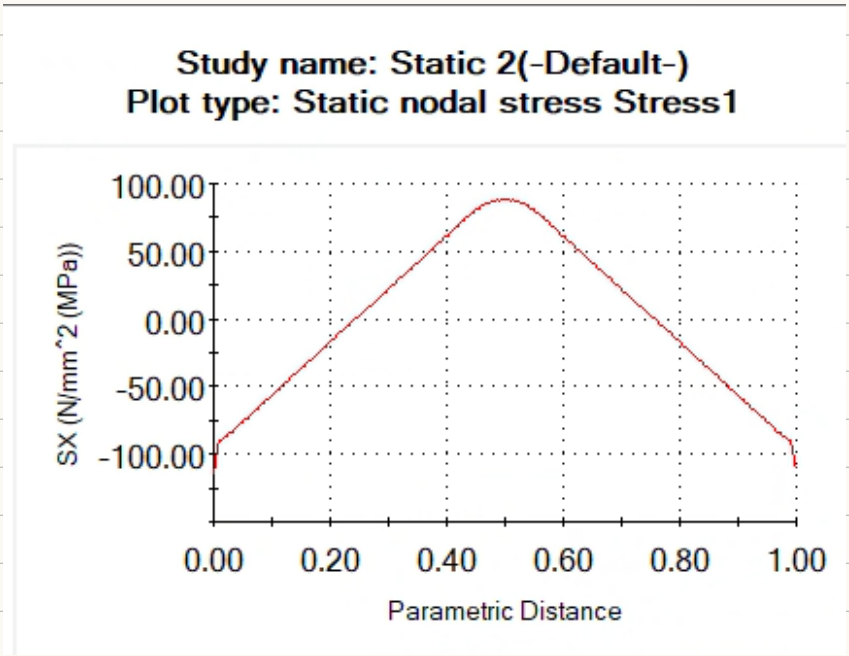
$M_1 = \frac{F}{2}x_1 - \frac{FL}{8}$

$0 = \frac{F}{2}x_1 - \frac{FL}{8}$

$x_1 = \frac{FL}{8} / \frac{F}{2} = \frac{L}{4}$

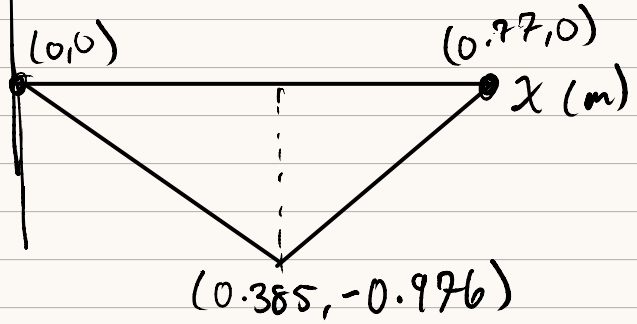
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l) cont.)



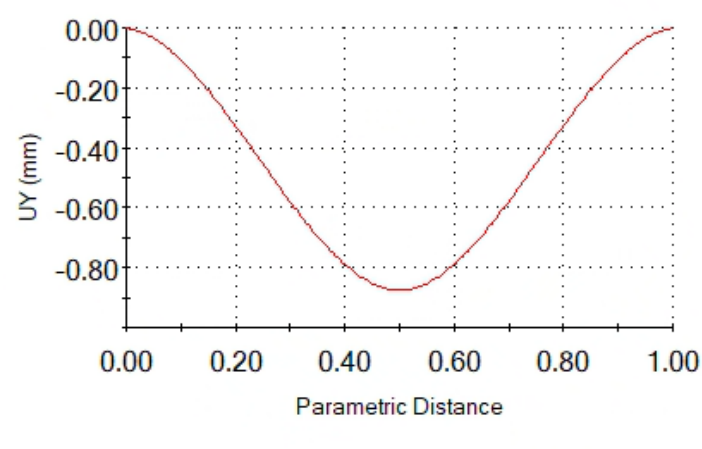
Between the two figures, the one from the ME10 lab has curve near the beginning, at the midpoint, and at the end. Although both share the same shape and very similar #s, the graph from the handwritten calculations are more sharp. The sharpness and curvature comes from the nodes. In our calculations, we only had one point, while our ME10 lab had multiple of nodes, creating the curve.

m) deflection (mm)



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m) cont)



Similar to part l, they have the same shape, where they follow the same trend. However, the FEA model from MEIO it curves down to up from the midpoint. At the beginning, it curves down and near the end, it starts to curve down at a certain point. For the handwritten calculations, it has a sharp turn from going down to going up.

Conclusion)

In this project, there were a lot of things to know in order to complete it. The main takeaway is that the computer may not be as exact as we thought they would be compared to our handwritten calculations. The values of both of them differ from one another. We also learned about indeterminant scenarios and how to deal with them step by step, which was learned in the lecture. The fun part about this is that we are visualizing why knowing the math is important for our careers. Therefore, it showed us to be cautious of the calculations made online and to ensure that they are correct by doing hand calculations.

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